

Name: Nguyễn Văn Đức

Student ID: 11247153

Class: DS66B

Lab 1: Film Ranking (The Basics)

```
1  SELECT
2      f.film_id,
3      f.title,
4      c.name AS category,
5      f.length,
6      DENSE_RANK() OVER (
7          PARTITION BY c.name
8          ORDER BY f.length DESC
9      ) AS length_rank
10 FROM film AS f
11 JOIN film_category AS fc
12     ON f.film_id = fc.film_id
13 JOIN category AS c
14     ON fc.category_id = c.category_id;
15
```

Data Output Messages Notifications

	film_id integer	title character varying (255)	category character varying (25)	length smallint	length_rank bigint
1	991	Worst Banger	Action	185	1
2	212	Darn Forrester	Action	185	1
3	126	Casualties Encino	Action	179	2
4	707	Quest Mussolini	Action	177	3
5	287	Entrapment Satisfaction	Action	176	4
6	511	Lawrence Love	Action	175	5
7	927	Uprising Uptown	Action	174	6
8	549	Magnolia Forrester	Action	171	7
9	250	Dragon Squad	Action	170	8
10	29	Antitrust Tomatoes	Action	168	9
11	115	Campus Remember	Action	167	10
12	825	Speakeasy Date	Action	165	11
13	850	Story Side	Action	163	12
14	327	Fool Mockingbird	Action	158	13
15	579	Minds Truman	Action	149	14
16	271	Easy Gladiator	Action	148	15

Lab 2: Growth Analysis (Time Series)

```
1  WITH monthly_revenue AS (  
2      SELECT  
3          DATE_TRUNC('month', payment_date) AS month,  
4          SUM(amount) AS revenue  
5      FROM payment  
6      GROUP BY DATE_TRUNC('month', payment_date)  
7  ),  
8  monthly_with_previous AS (  
9      SELECT  
10         month,  
11         revenue,  
12         LAG(revenue) OVER (  
13             ORDER BY month  
14         ) AS previous_revenue  
15     FROM monthly_revenue  
16 )  
17 SELECT  
18     month,  
19     revenue,  
20     previous_revenue,  
21     (revenue - previous_revenue)  
22         / previous_revenue * 100 AS growth_pct  
23 FROM monthly_with_previous  
24 ORDER BY month;
```

Data Output Messages Notifications

	month timestamp without time zone	revenue numeric	previous_revenue numeric	growth_pct numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

Lab 3: Customer Lifetime Value (Running Total)

```
1  SELECT
2      payment_id,
3      customer_id,
4      payment_date,
5      amount,
6      SUM(amount) OVER (
7          PARTITION BY customer_id
8          ORDER BY payment_date
9      ) AS cumulative_spent
10 FROM payment
11 ORDER BY customer_id, payment_date;
```

Data Output Messages Notifications

	payment_id [PK] integer	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_spent numeric
1	18495	1	2007-02-14 23:22:38.996577	5.99	5.99
2	18496	1	2007-02-15 16:31:19.996577	0.99	6.98
3	18497	1	2007-02-15 19:37:12.996577	9.99	16.97
4	18498	1	2007-02-16 13:47:23.996577	4.99	21.96
5	18499	1	2007-02-18 07:10:14.996577	4.99	26.95
6	18500	1	2007-02-18 12:02:25.996577	0.99	27.94
7	18501	1	2007-02-21 04:53:11.996577	3.99	31.93
8	22680	1	2007-03-01 07:19:30.996577	4.99	36.92
9	22681	1	2007-03-02 14:05:18.996577	3.99	40.91
10	22682	1	2007-03-02 16:30:04.996577	0.99	41.90
11	22683	1	2007-03-17 11:06:20.996577	4.99	46.89
12	22684	1	2007-03-18 02:25:55.996577	0.99	47.88
13	22685	1	2007-03-19 08:23:42.996577	0.99	48.87
14	22686	1	2007-03-19 12:25:20.996577	2.99	51.86

Lab 4: The "Top-N" Filter Challenge

```

1  WITH customer_spend AS (
2      SELECT c.customer_id, co.country, SUM(p.amount) AS total_spend
3      FROM customer c
4      JOIN payment p ON c.customer_id = p.customer_id
5      JOIN address a ON c.address_id = a.address_id
6      JOIN city ci ON a.city_id = ci.city_id
7      JOIN country co ON ci.country_id = co.country_id
8      GROUP BY c.customer_id, co.country
9  ),
10 ranked_customers AS (
11     SELECT customer_id, country, total_spend,
12            RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) AS spend_rank
13     FROM customer_spend
14 )
15 SELECT *
16 FROM ranked_customers
17 WHERE spend_rank = 1
18 ORDER BY country;

```

Data Output Messages Notifications

SQL Sho

	customer_id integer	country character varying (50)	total_spend numeric	spend_rank bigint
1	218	Afghanistan	67.82	1
2	176	Algeria	151.68	1
3	320	American Samoa	47.85	1
4	528	Angola	93.80	1
5	381	Anguilla	99.68	1
6	560	Argentina	129.71	1
7	41	Armenia	118.75	1
8	266	Austria	113.74	1
9	334	Azerbaijan	108.75	1
10	590	Bahrain	108.76	1
11	21	Bangladesh	146.68	1
12	144	Belarus	189.60	1
13	431	Bolivia	92.78	1
14	178	Brazil	194.61	1